

Outline

01. Introduction to phylogenetics
02. Gene family evolution
03. The anatomy of a tree
04. How to build a tree

Outline

- 01. Introduction to phylogenetics**
- 02. Gene family evolution
- 03. The anatomy of a tree
- 04. How to build a tree

Why do we study phylogeny

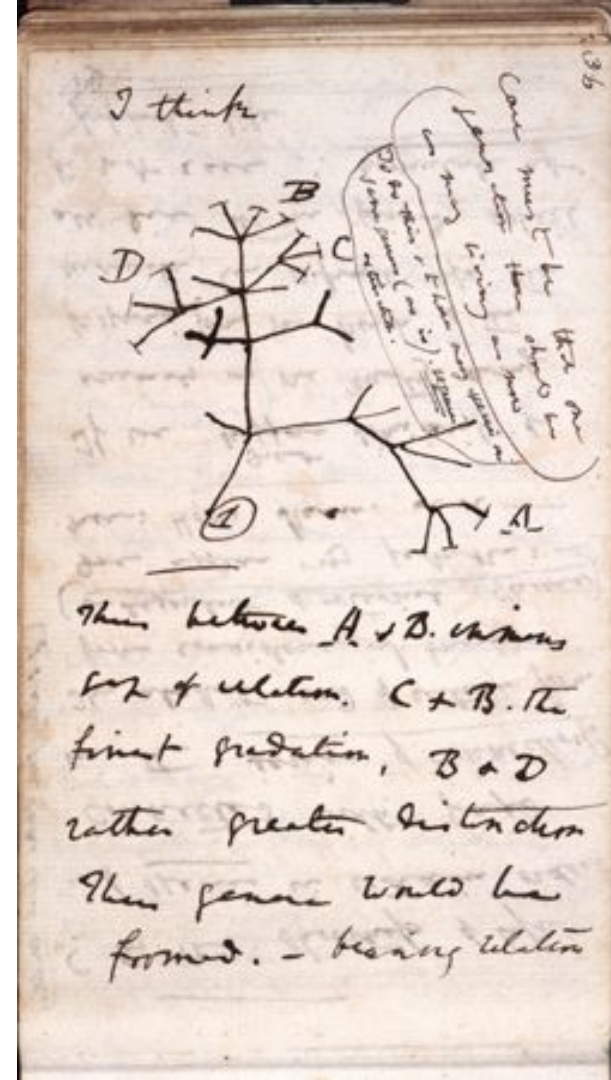
How do genes
change over time?

How are species
related to each
other?

Where do new
traits come from?

The tree of life

- The concept of representing species in a tree, with species at the tips converging back to shared common ancestry is old
- One goal of evolutionary biology is to understand the relationships and processes that underlie a tree
- Understanding the process of molecular evolution is essential to understand how genes function, and is directly applicable to the fields of epidemiology and medicine

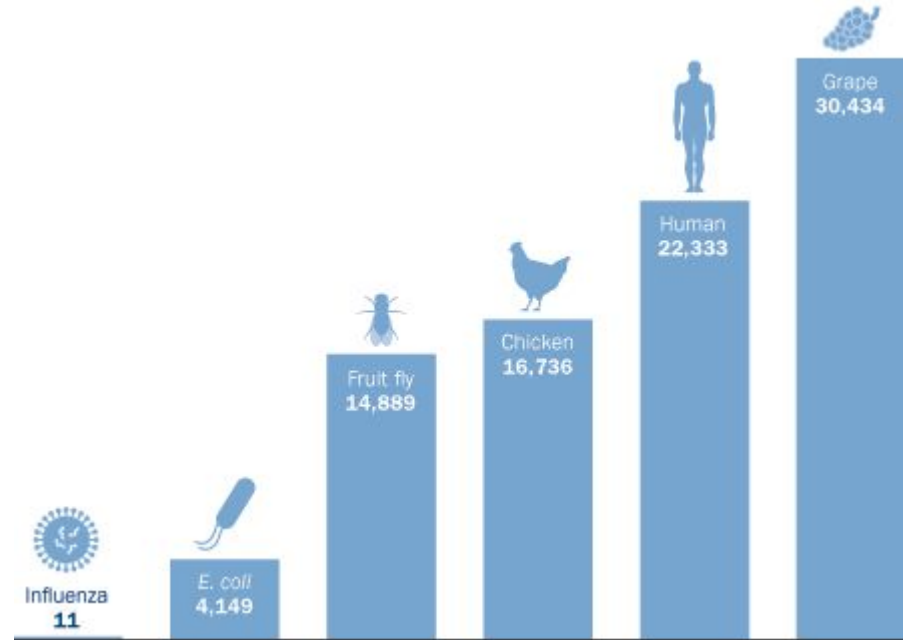


Outline

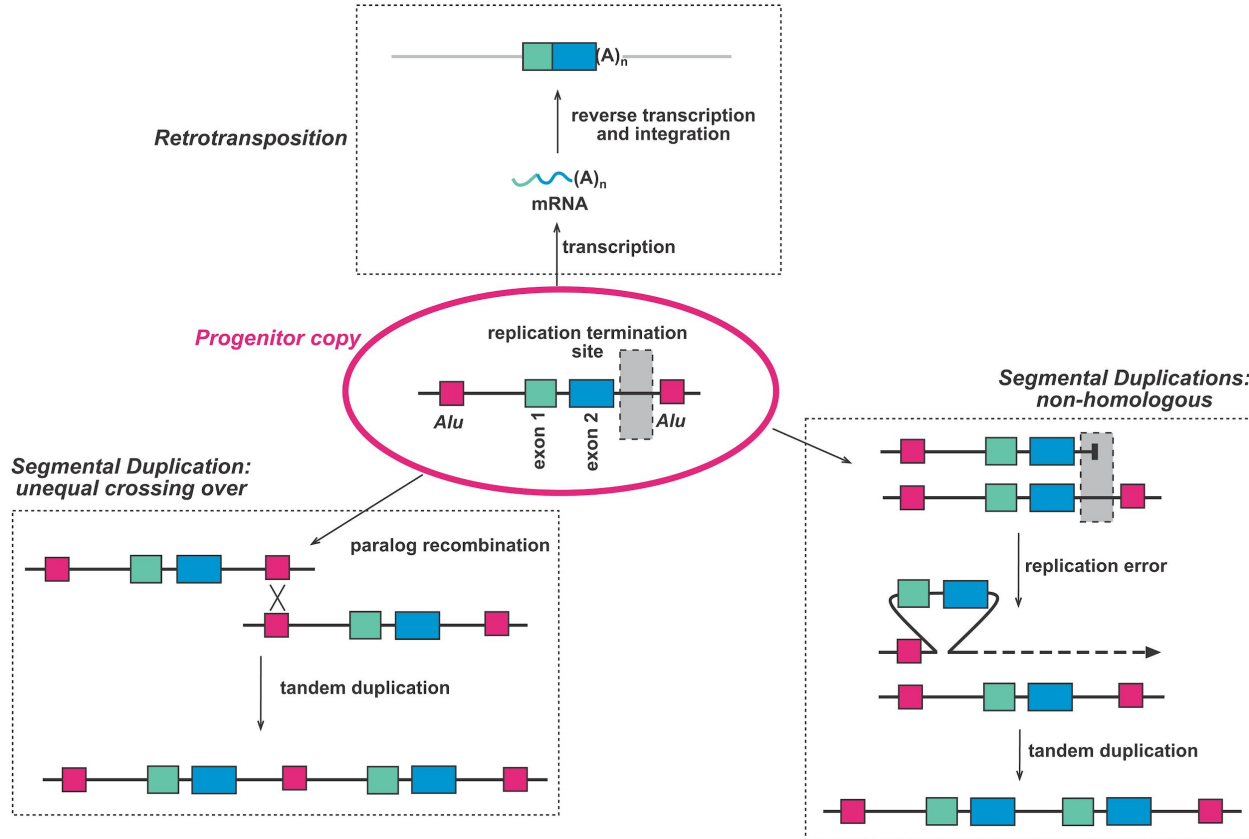
- 01. Introduction to phylogenetics
- 02. Gene family evolution**
- 03. The anatomy of a tree
- 04. How to build a tree

Where do genes come from?

- The number of protein coding genes encoded in the genome can vary dramatically between species

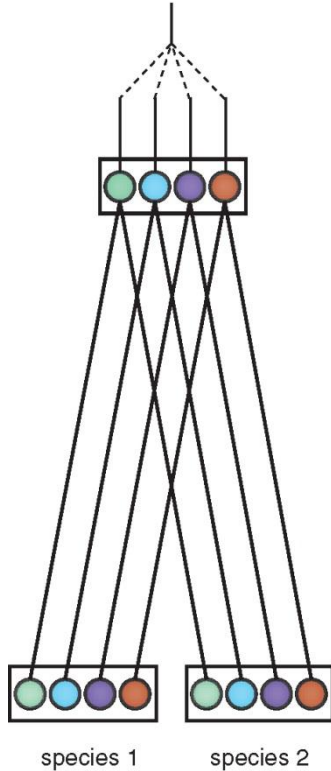


Mechanisms of gene duplication

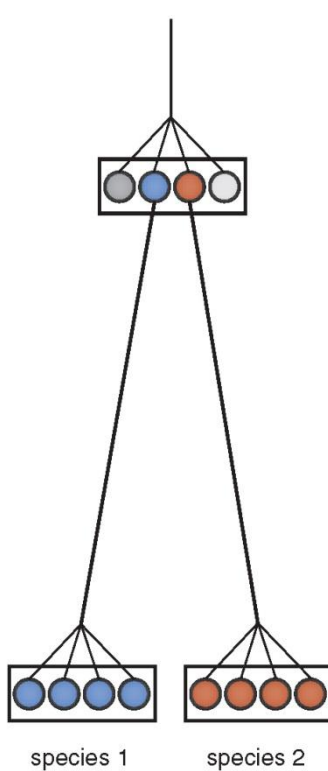


What is the fate of duplicate genes?

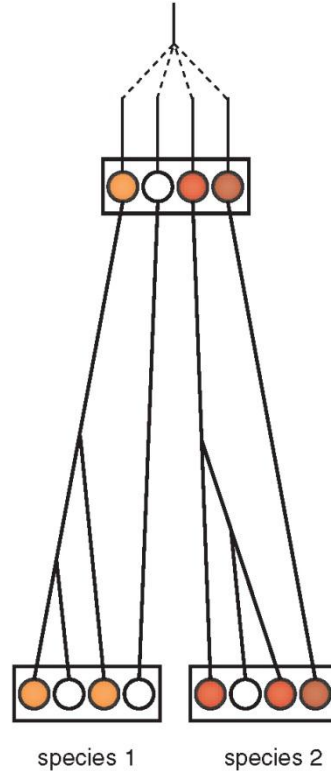
(a) Divergent evolution



(b) Concerted evolution

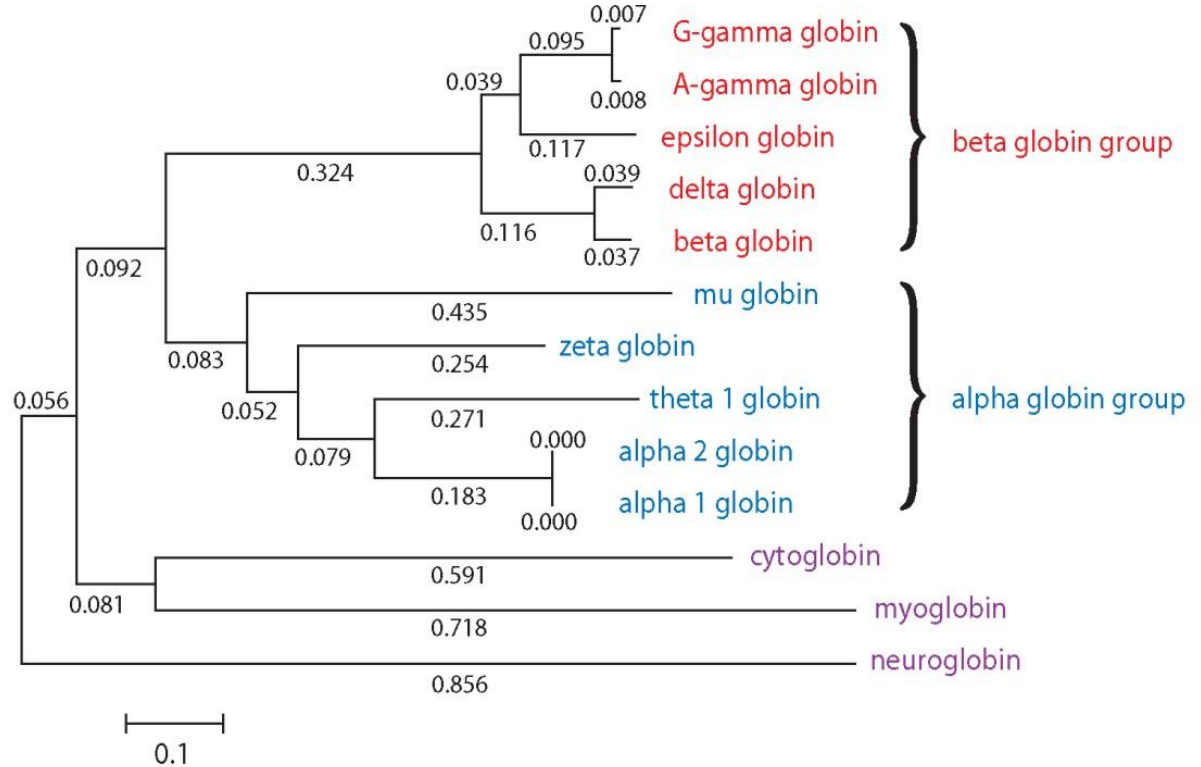
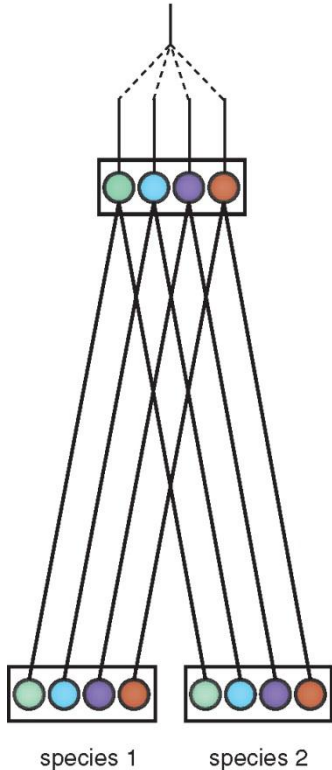


(c) Birth-and-death evolution



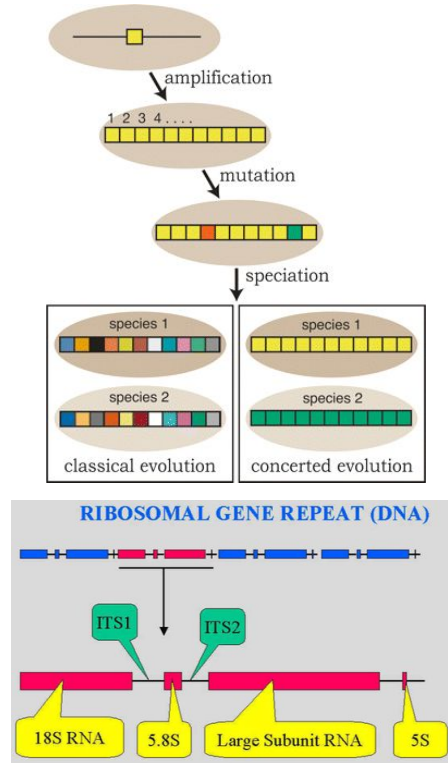
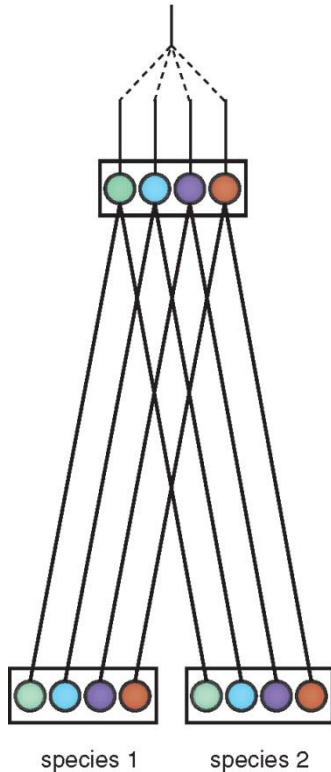
Divergent evolution of duplicates

(a) Divergent evolution

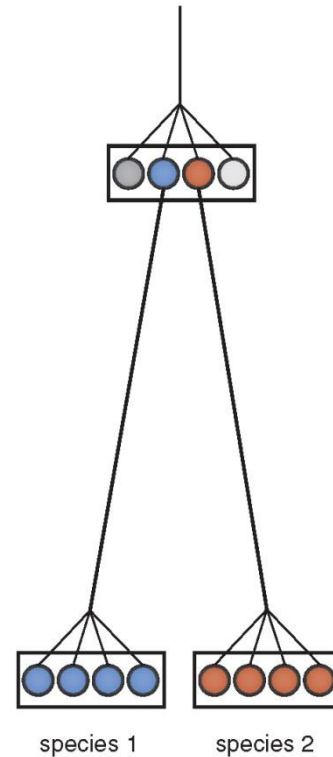


Concerted evolution of duplicates

(a) Divergent evolution

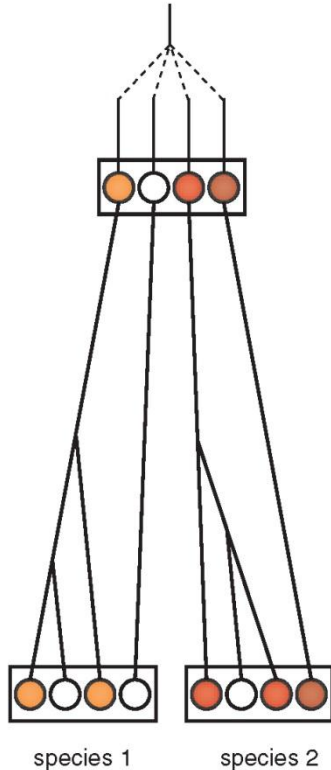


(b) Concerted evolution



Evolution by birth-death processes

(c) Birth-and-death evolution



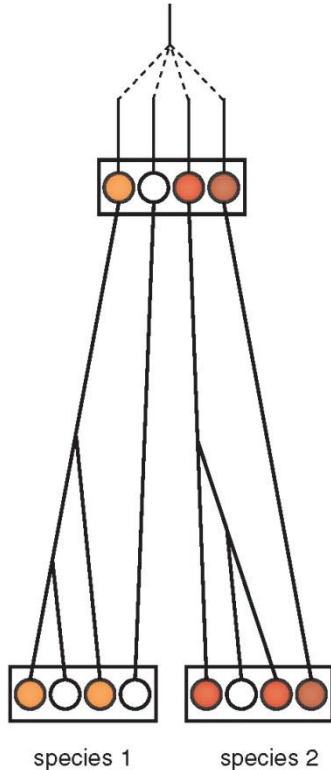
Unprocessed pseudogenes:

-Originally functional, but have acquired mutations that render it unable to function

Pseudogene	TATA box	Initiation codon	Frame-shift	Pre-mature stop	Essential amino acid	Splice GT/AG rule	Altered stop codon	Poly-adenylation signal
Human $\psi\alpha 1$		X	X	X	X	X	X	X
Human $\psi\zeta 1$				X				
Mouse $\psi\alpha 3$	X		X	X		X		
Mouse $\psi\alpha 4$			X		X			
Mouse $\beta h 3$	X	X	X	X	X	X	X	X
Goat $\psi\beta^x$	X		X	X	X	X	X	X
Goat $\psi\beta^z$	X		X	X	X	X	X	X
Rabbit $\psi\beta 2$			X	X	X	X		

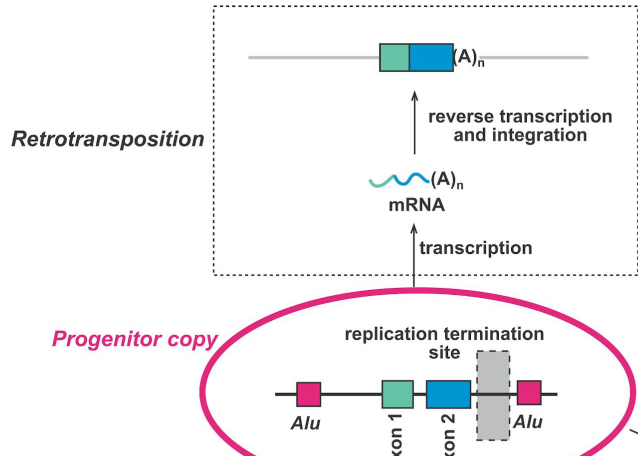
Evolution by birth-death processes

(c) Birth-and-death evolution



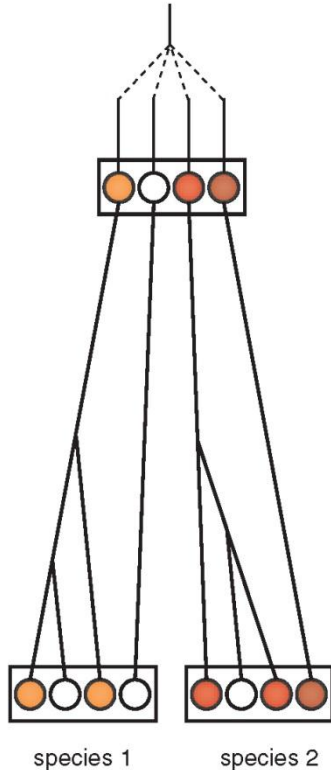
Processed pseudogenes:

-Probably never functional, quickly accumulate mutations

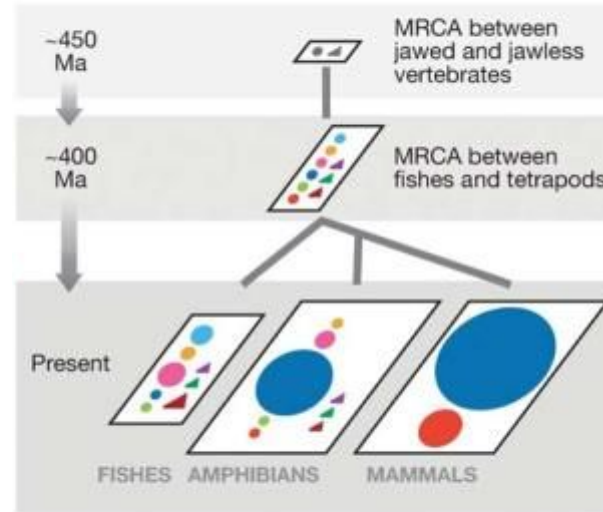


Evolution by birth-death processes

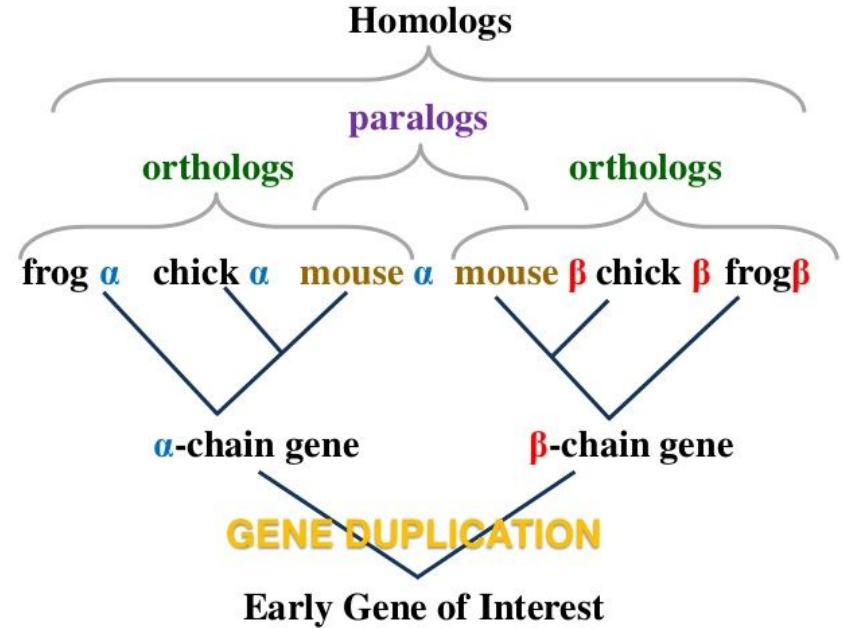
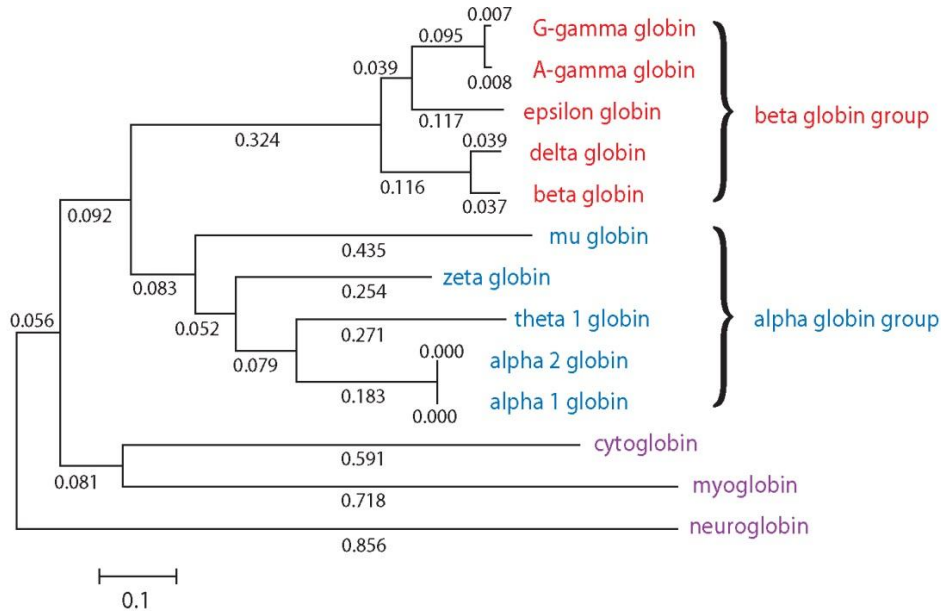
(c) Birth-and-death evolution



Some gene families show an extraordinary rate of birth and death: example odorant receptors



Homologs, paralogs and orthologs



- Orthologs- Homologs that arose from an ancestral gene by speciation
- Paralogs- Homologs that arose from a duplication within a genome

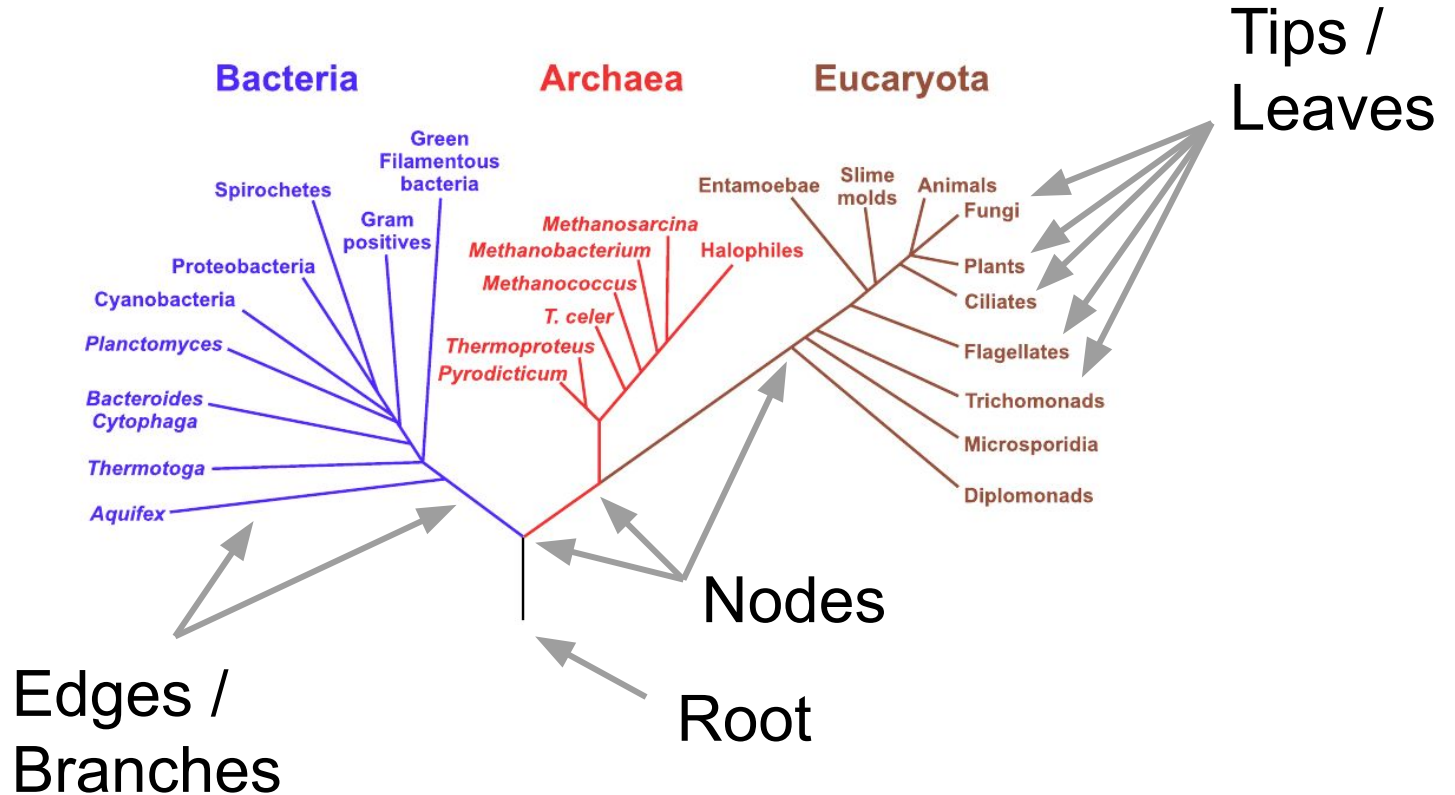
Summary

- New genes are derived from duplications of old genes constantly during evolution
- Some duplications are retained, though many are lost
- This can result in the expansion of gene families
- Expansion of gene families can sometimes be adaptive

Outline

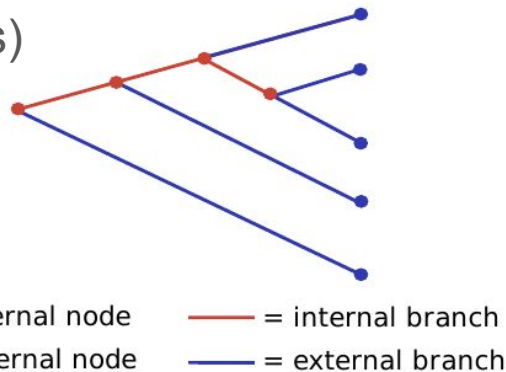
- 01. Introduction to phylogenetics
- 02. Gene family evolution
- 03. The anatomy of a tree**
- 04. How to build a tree

The anatomy of a tree



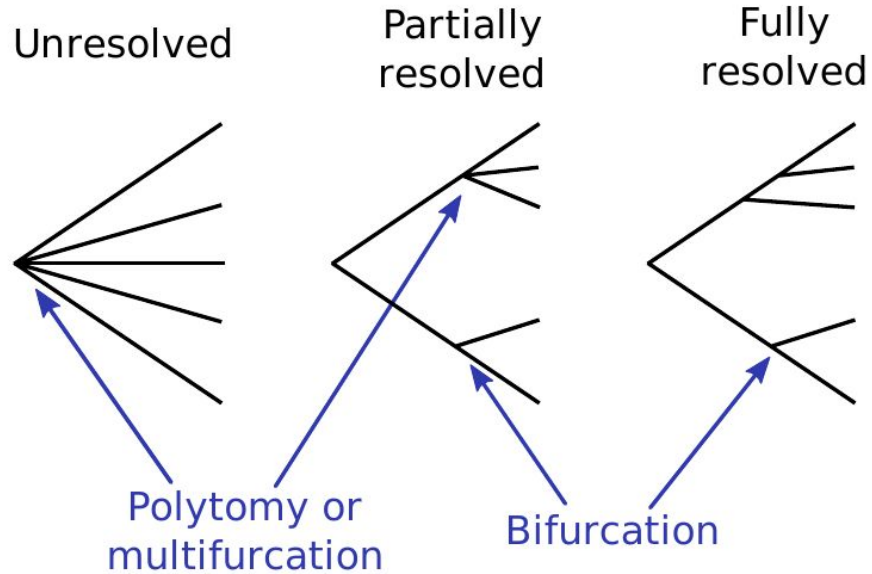
Tree Basics

- Typically, binary trees are used in phylogenetics, where three branches meet at each branch node.
 - Tree components
 - internal and external branches (edges)
 - internal and external nodes
- 



- A true phylogeny has a root, which represents the ultimate ancestor for all the other items in a tree.
- Certain algorithms like parsimony or probabilistic models provide no information about the position of the root in a tree.

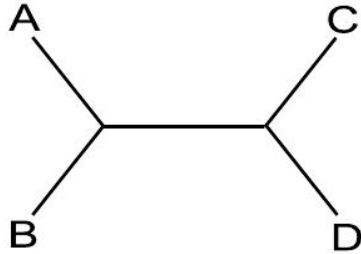
Resolution of a tree



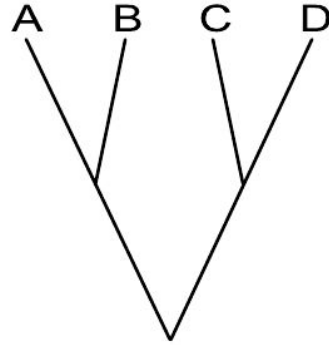
Some parts of the tree of life are fully resolved, others are only partially resolved or completely unresolved.

Rooted and unrooted trees

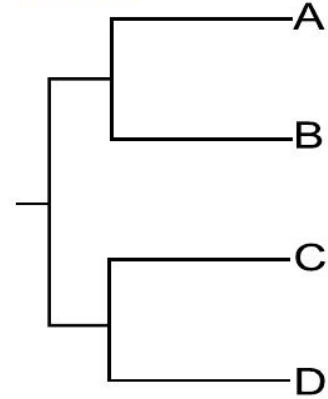
Unrooted Tree



Rooted Trees



Slanted
style

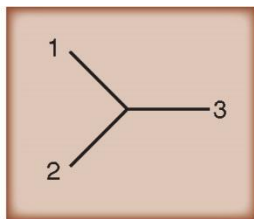


Rectangular
style

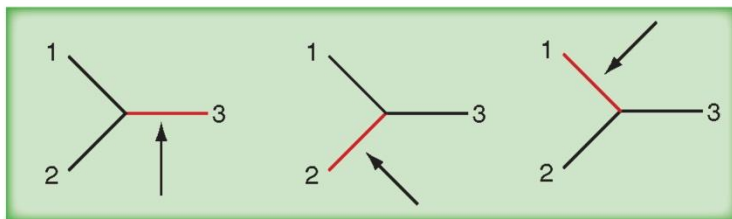
Newick format: ((A:0.5, B:0.5):0.1, (C:0.5, D:0.5):0.1);

Possible root positioning

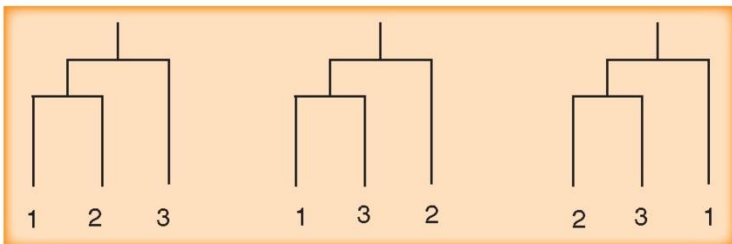
(a)



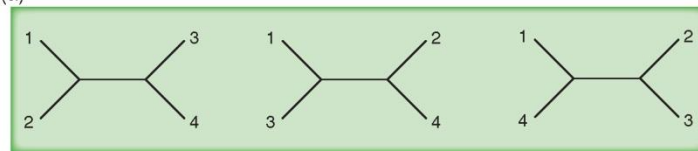
(b)



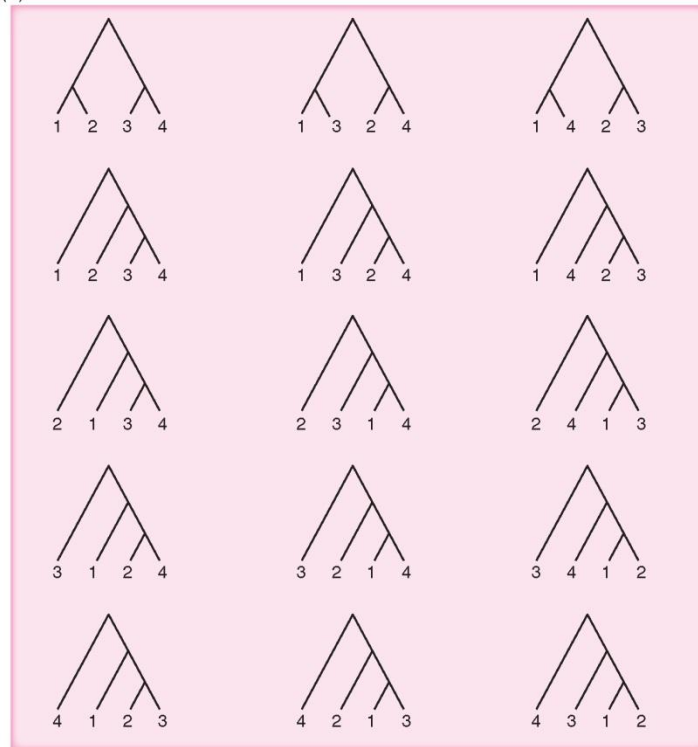
(c)



(a)

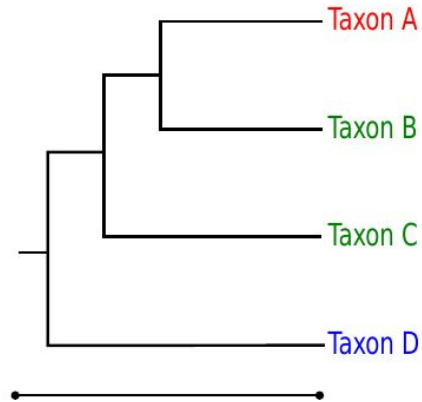


(b)



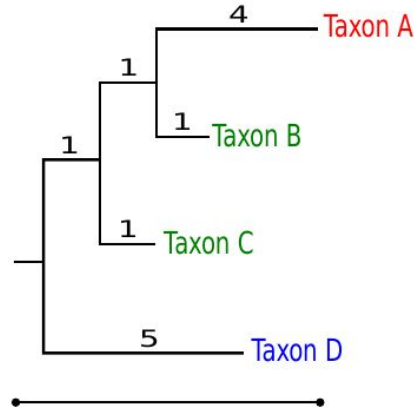
The meaning of branch lengths

Cladogram



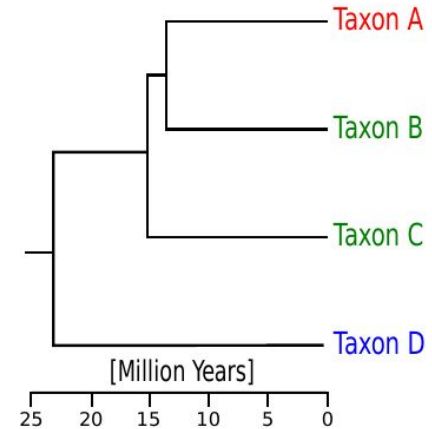
Branch lengths have no meaning.

Phylogram



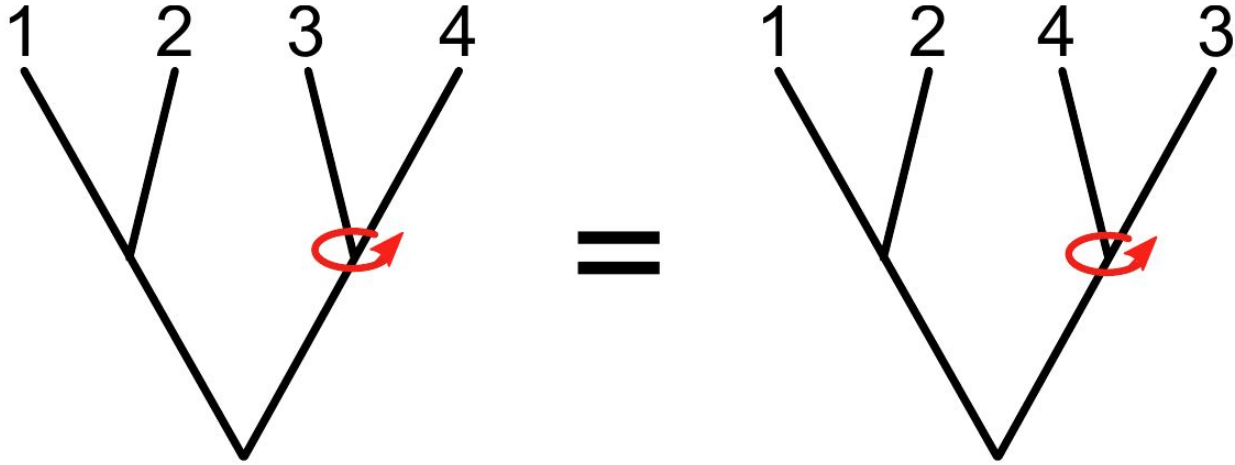
Branch lengths are proportional to (genetic) change.

Ultrametric Tree



Branch lengths are proportional to time.

Rotation of branches



Rotations of internal nodes yield exactly the same tree.

Outline

- 01. Introduction to phylogenetics
- 02. Gene family evolution
- 03. The molecular clock
- 04. The anatomy of a tree
- 05. How to build a tree**

How to evaluate a tree building method

- **Consistency:** will the method converge on the correct solution given enough data?
- **Efficiency:** how fast is the method?
- **Robustness:** will minor violations of the assumptions result in poor estimates of phylogeny?

Trait evolution and phylogeny

Trees can help us understand the order and timing of trait evolution

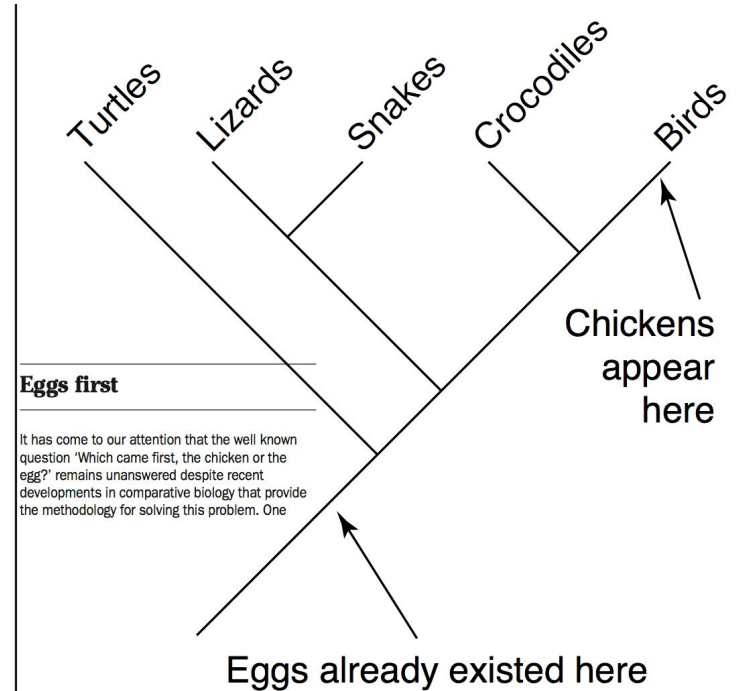


Fig. 1. Summarized phylogeny (based on the classification of Romer²) of the reptiles and birds showing the phylogenetic position of chickens and the presence of eggs.

Trait evolution and phylogeny: Parsimony

Group taxa by traits

	Vertebrae?	Bony skeleton?	Four limbs?	Amniotic egg?	Hair?	Two post-orbital fenestrae?
Sharks and relatives	YES	no	no	no	no	no
Ray-finned fishes	YES	YES	no	no	no	no
Amphibians	YES	YES	YES	no	no	no
Primates	YES	YES	YES	YES	YES	no
Rodents and rabbits	YES	YES	YES	YES	YES	no
Crocodiles and relatives	YES	YES	YES	YES	no	YES
Dinosaurs and birds	YES	YES	YES	YES	no	YES

*amniotic egg:
an egg in which the
embryo is surrounded
by the moisture-retaining
amnion membrane



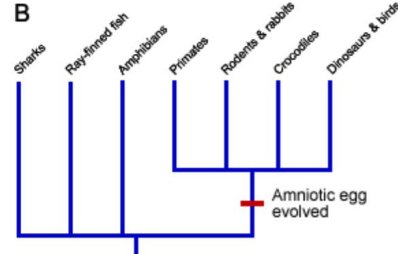
**post-orbital fenestrae:
holes in the skull
behind the eye



Group taxa by traits

A

	Vertebrae?	Bony skeleton?	Four limbs?	Amniotic egg?	Hair?	Two post-orbital fenestrae?
Sharks and relatives	YES	no	no	no	no	no
Ray-finned fishes	YES	YES	no	no	no	no
Amphibians	YES	YES	YES	no	no	no
Primates	YES	YES	YES	YES	YES	no
Rodents and rabbits	YES	YES	YES	YES	YES	no
Crocodiles and relatives	YES	YES	YES	YES	no	YES
Dinosaurs and birds	YES	YES	YES	YES	no	YES

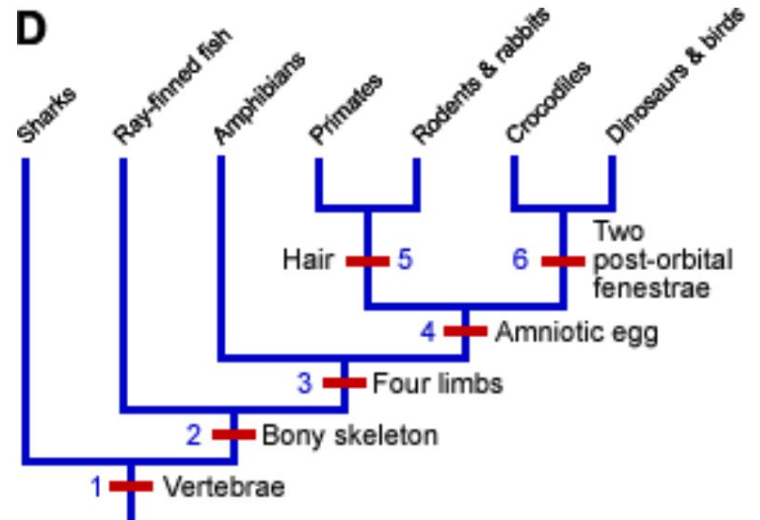


Trait evolution and phylogeny: Parsimony

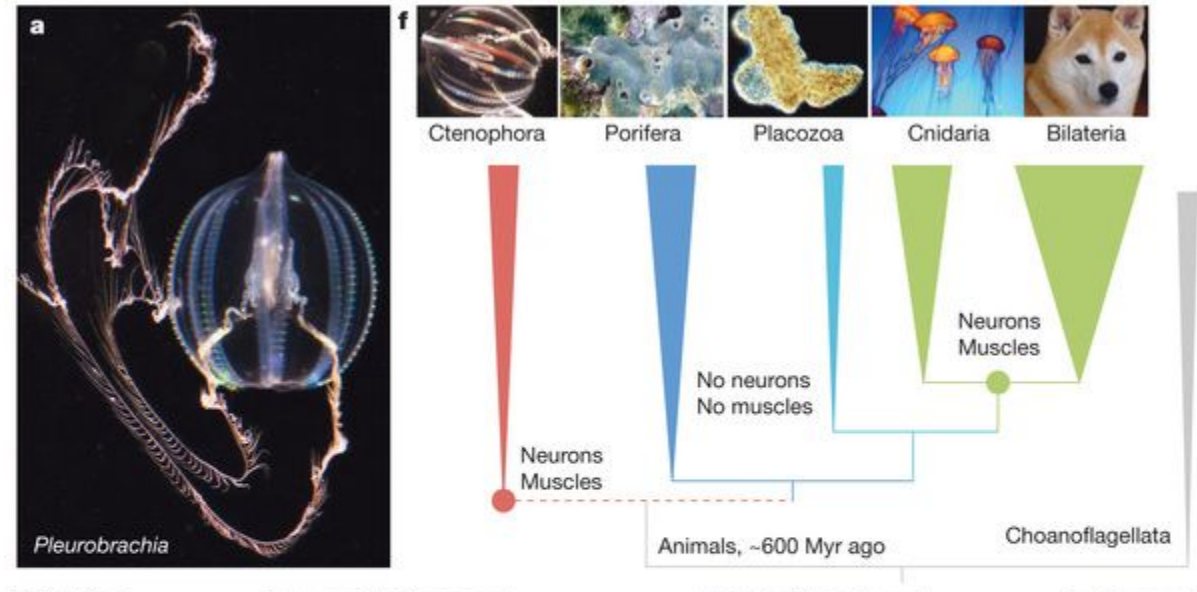
C

	Vertebrae?	Bony skeleton?	Four limbs?	Amniotic egg?	Hair?	Two post-orbital fenestrae?
Sharks and relatives	YES	no	no	no	no	no
Ray-finned fishes	YES	YES	no	no	no	no
Amphibians	YES	YES	YES	no	no	no
Primates	YES	YES	YES	YES	YES	no
Rodents and rabbits	YES	YES	YES	YES	YES	no
Crocodiles and relatives	YES	YES	YES	YES	no	YES
Dinosaurs and birds	YES	YES	YES	YES	no	YES
	1	2	3	4	5	6

D



Neuronal development in animals



Parsimony methods and sequence data

- Basic principle: find the tree that explains the observed sequences with the minimal number of substitutions.
- Parsimony assigns a cost to a given tree.
- This requires searching through all possible trees or a subset of trees that contains the best or close to best tree topology.

Applying parsimony to sequences

Steps:

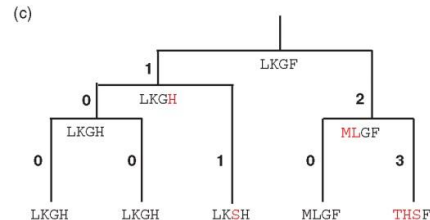
1. Find informative sites
2. Construct all (or a subset of) possible trees
3. Count the number of changes on the tree (the cost) and select the tree with the lowest cost

(a)

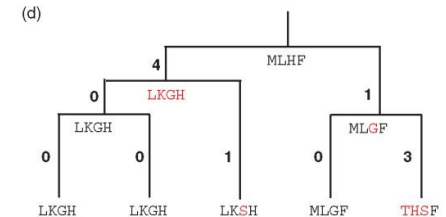
(b)

```

kangaroo  LKGF
porpoise  LKGF
gray seal LKSH
horse α   MLGF
kangaroo α THSF
    
```



Total cost: 7



Total cost: 9

Distance based clustering methods

- Algorithmic methods in which the algorithm itself defines the tree selection criterion.
- No optimality criteria applied (measure of fit to the data).
- Advantage: tend to be very fast (efficient) computations that produce singular trees.
- Disadvantages:
 - Do not allow evaluation of competing hypotheses.
 - No objective function (e.g. likelihood, number of steps) is used to compare different trees to each other, even if numerous other trees could explain the data equally well.
- Examples that construct trees from distances:
 - UPGMA
 - Neighbor joining (NJ)

Distance matrix construction

- DNA Alignment

G1 TTATTAA

G2 AATTTAA

G3 AAAAATA

G4 AAAAAAT

- Distance Matrix

	G1	G2	G3	G4
G1		0.43	0.71	0.71
G2	3		0.57	0.57
G3	5	4		0.29
G4	5	4	2	

Absolute distances in bottom triangle and uncorrected relative distances in top triangle.

Maximum likelihood

Explicitly models the likelihood of a tree given a set of data

- allows for varied rates of evolution among branches and sites
- Often produces the most accurate trees assuming the parameters are correctly specified
- very computationally intensive, making it difficult to implement for large numbers of species

Positioning the root

Outgroup Method

Rooting by including one or more outgroup taxa/sequences.

Gene Duplication

Paralogous gene duplication predating the common ancestor of a clade are used.

Midpoint Rooting

Tree is rooted by midpoint between the two most distant branches.

Bootstrapping

- Data points are randomly resampled from the original data set, with replacement, until new data sets with the original number of observations are obtained.
- Statistic of interest (*e.g.* a tree) is computed for each replicated data set.
- Agreement among the resulting trees is summarized with a majority-rule consensus tree (agreement > 50%).
- A bootstrap proportion (BP) is the frequency of occurrence of a clade (for all replicated data sets) and is a measure of support for a group.

Bootstrapping example

[A] Original multiple alignment (MSA)

	1	2	3	4	5	6	7	8
A	C	C	S	S	S	S	S	S
B	C	C	P	P	P	P	P	P
C	C	C	C	C	C	P	P	P
D	P	P	C	C	C	C	C	C
Outgroup	C	C	C	C	C	C	C	C

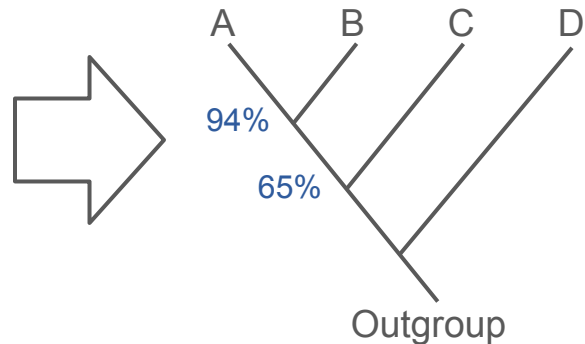
[B] Resampling of MSA (example)

	1	2	2	4	4	6	6	8
A	C	C	C	S	S	S	S	S
B	C	C	C	P	P	P	P	P
C	C	C	C	C	C	P	P	P
D	P	P	P	C	C	C	C	C
Outgroup	C	C	C	C	C	C	C	C

[B] Resampling of MSA (continued)

- Compute 100, 1000 or more resampled MSAs
- Compute for original MSA and each resampled MSAs a tree
- Determine how often each clade (sub-cluster) in original tree occurs in trees of resampled MSAs according to a majority rule.
- Frequencies/percentages are reported as **bootstrap values** at corresponding tree branches.

[C] Majority rule tree with bootstrap proportions (BP) at branch nodes



How to interpret bootstraps

- Bootstrap proportions are sometimes interpreted as confidence intervals for phylogenies.
- This interpretation assumes that characters (e.g. nucleotide or amino acid sites) are independent and identically distributed, and that the method is consistent.
- Even in a best case scenario bootstrapping sometimes gives underestimates of accuracy at high bootstrap values and overestimates of accuracy at low bootstrap values.

Steps of building a phylogenetic tree

1. Select the Appropriate Sequences for a Phylogenetic Question

Important: sequences should show significant similarity.

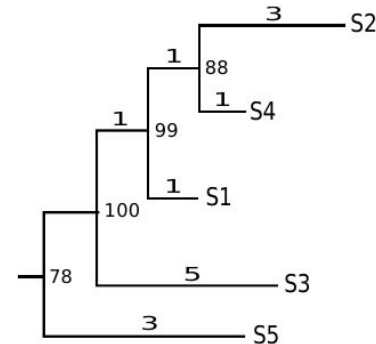
2. Create a Multiple Alignment for Chosen Sequences

Important: unalignable sequence areas should be removed.

S1	FM	PFSACKRI	CAGEGLARMELFLFLT	450
S2	FM	PFSAGKRICV	GEALAGMELFLFLT	450
S3	.LA	FGCGARVCL	GEPLARLELVVLT	443
S4	SL	PFGFGKRSCM	GRRRLAELELQMALA	470
S5	YT	PFGSGPRNCI	GMRFALNMKIALI	457
consensus	.	PFG.GkR.C	Ge.LA.mELf.l.Lt	

3. Compute a Distance Matrix for Multiple Alignment

	S1	S2	S3	S4	S5
S1	0.0	0.43	0.71	0.71	0.48
S2		0.0	0.57	0.57	0.39
S3			0.0	0.29	0.21
S4				0.0	0.13
S5					0.0



4. Calculate Phylogenetic Tree

Important: choose a tree building method.

5. Tree Post Processing

Important: tree rooting and bootstrapping.